

Advanced (Habanero)

- Q1.** The vertex of the parabola $y = x^2 + 4x + 4$ is
- (A) $(0, 0)$
(B) $(-2, 0)$
(C) $(-2, 4)$
(D) $(0, 4)$
(E) $(-4, 0)$
- Q2.** The axis of symmetry of the parabola $y = x^2 - 6x + 8$ is
- (A) $x = -2$
(B) $x = 2$
(C) $x = 3$
(D) $x = 5$
(E) $x = 3$
- Q3.** The y -intercept of the parabola $y = x^2 - 2x - 3$ is
- (A) $(-3, 0)$
(B) $(0, -3)$
(C) $(3, 0)$
(D) $(-2, 0)$
(E) $(2, 0)$
- Q4.** The direction of the parabola $y = -x^2 + 2x + 1$ is
- (A) upward
(B) downward
(C) leftward
(D) rightward
(E) horizontal
- Q5.** The maximum value of the parabola $y = -x^2 + 4x - 3$ occurs at
- (A) $x = -2$
(B) $x = 0$
(C) $x = 2$
(D) $x = 4$
(E) $x = 2$
- Q6.** The x -intercepts of the parabola $y = x^2 + 2x + 1$ are
- (A) $(-1, 0)$ and $(1, 0)$
(B) $(-1, 0)$ and $(-1, 0)$
(C) $(-2, 0)$ and $(0, 0)$
(D) $(0, 0)$ and $(0, 0)$
(E) $(-1, 0)$
- Q7.** The equation of the axis of symmetry of the parabola $y = x^2 + 4x + 4$ is
- (A) $x = -2$
(B) $x = 0$
(C) $x = 2$
(D) $x = -4$
(E) $x = -1$
- Q8.** Which of the following parabolas has a minimum value?
- (A) $y = x^2 + 2x + 1$
(B) $y = -x^2 + 2x + 1$
(C) $y = x^2 - 2x + 1$
(D) $y = -x^2 - 2x - 1$
(E) $y = x^2 + 1$

Open Ended Questions (FRQ)

- Q9.** Graph the parabola $y = x^2 + 2x - 3$ and find its vertex, axis of symmetry, x -intercepts, and y -intercept.
- Q10.** Find the maximum or minimum value of the parabola $y = -x^2 + 4x - 3$. Justify your answer.

Chef's Tips

- Tip 1: Make sure to read the questions carefully and choose the correct answer.
- Tip 2: Use graphing calculators or software to visualize the parabolas and check your answers.
- Tip 3: Check your work and use the properties of parabolas to justify your answers.